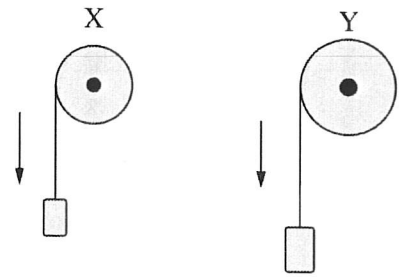


B

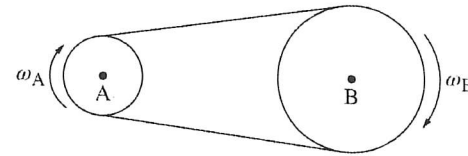
1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration a . As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity ω_Y of Pulley Y compare to the final angular velocity ω_X of Pulley X?



- (A) $\omega_Y > \omega_X$
- (B) $\omega_Y < \omega_X$
- (C) $\omega_Y = \omega_X$
- (D) It cannot be determined without knowing the relative masses of each block.

C

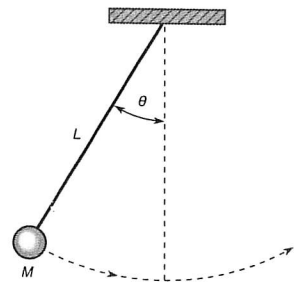
2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed ω_A and Wheel B has constant angular speed ω_B . Which of the following correctly relates ω_A



- (A) $\omega_A = \frac{\omega_B}{3}$
- (B) $\omega_A = \omega_B$
- (C) $\omega_A = 3\omega_B$
- (D) $\omega_A = 6\omega_B$

C

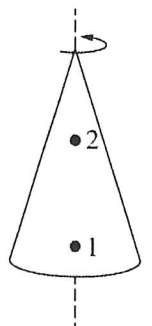
3. A pendulum is formed from a mass M connected to a string of length L . The pendulum is initially raised to form an angle θ (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be T (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?



- (A) $\frac{\theta}{T}$
- (B) $\frac{\theta L}{T}$
- (C) $\frac{2\theta L}{T}$
- (D) $\frac{4\theta L}{T}$

C

4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?



- (A) The angular speed of the insect is greater at Point 1 than at Point 2.
- (B) The angular speed of the insect is greater at Point 2 than at Point 1.
- (C) The linear speed of the insect is greater at Point 1 than at Point 2.
- (D) The linear speed of the insect is greater at Point 2 than at Point 1.

B Side

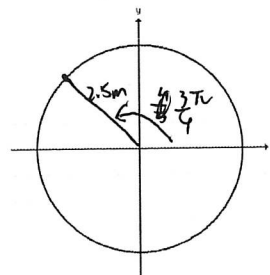
1. A disk has a radius of 2.5 m and undergoes an angular displacement of $3\pi/4$ radians.

- a. Sketch this angular displacement
b. Convert this measurement to degrees.

$$\frac{3}{4}\pi = 135^\circ$$

- c. Determine the arc length formed.

$$2.5 \times \frac{3}{4}\pi = \frac{7.5}{4}\pi$$



2. A woman walks in a circular path of radius 20 m. The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

- a. Sketch this angular displacement
b. Convert her angular displacement to radians.

$$2.2 \text{ rotations} = \frac{22}{5}\pi$$

- c. Calculate the woman's linear displacement.

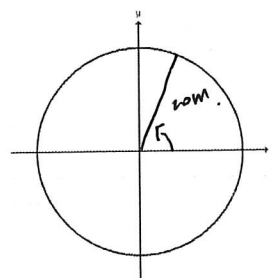
$$\frac{22}{5}\pi \times 20 = 88\pi$$

- d. Calculate the woman's angular velocity (in rad/s)

$$\frac{\frac{22\pi}{5}}{20} = \frac{11}{50}\pi \text{ rad/s}$$

- e. Calculate the woman's linear velocity (in m/s)

$$\frac{88\pi}{20} = 4.4\pi \text{ m/s}$$



3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

- a. Determine the radius of the record

$$r = \frac{1.3}{2\pi}$$

- b. Determine the angular velocity of the record (in deg/s)

$$\omega = \frac{1.3 \text{ m}}{1.2 \text{ s}} = \frac{1.3}{1.2} \text{ m/s}$$

$$W = \frac{2\pi}{1.2 \text{ s}} = \frac{2\pi}{1.2} = \frac{360^\circ}{1.2} = 300^\circ/\text{s}$$

